

Java Spring Boot 4 Backend – UML Class Diagram

Spring Boot 4.1.1 • Spring AI 2.0.1 Dual Models (Llama 3.3 & Mistral) • MongoDB Repository • Qdrant Vector Retrieval

PACKAGE: com.personallibrary.controller

**<<RestController>>
DocumentController**

- documentService: DocumentService
- extractionService: BibTeXExtractionService
- summarizationService: AiSummarizationService
- vectorRagService: VectorRagService
- storageService: StorageService

+ getDocuments(fileName, title, author, edition, format, content, page, pageSize, sortBy, sortOrder): ResponseEntity<PaginatedResponse>
+ getDocument(guid: String): ResponseEntity<DocumentResponse>
+ uploadDocument(file: MultipartFile, metadataJson: String): ResponseEntity<DocumentResponse>
+ overwriteDocumentVersion(guid: String, file: MultipartFile, metadata: String): ResponseEntity<DocumentResponse>
+ extractMetadata(payload: Map<String, String>): ResponseEntity<BibTeXMetadata>
+ deleteDocument(guid: String): ResponseEntity<Map<String, Object>>
+ regenerateSummary(guid: String, request: SummarizeRequest): ResponseEntity<SummaryRecord>
+ chatWithDocument(guid: String, request: ChatRequest): ResponseEntity<ChatResponse>

calls

PACKAGE: com.personallibrary.service

**<<Service>>
DocumentService**

- documentRepository: DocumentRepository
- storageService: StorageService
- extractionService: BibTeXExtractionService
- summarizationService: AiSummarizationService
- vectorRagService: VectorRagService

+ uploadDocument(file: MultipartFile, userBibtex: BibTeXMetadata): DocumentResponse
+ overwriteDocument(existingGuid: String, newFile: MultipartFile, updatedBibtex: BibTeXMetadata): DocumentResponse
+ searchDocuments(criteria: DocumentQueryCriteria, pageable: Pageable): PaginatedResponse<DocumentResponse>
+ getDocumentByGuid(guid: String): DocumentResponse

AiSummarizationService

- llamaChatClient: ChatClient
- mistralChatClient: ChatClient

+ generateDualSummaries(title, text, bibtex): Map
+ generateSummaryForModel(modelKey, title, text, bibtex): SummaryRecord

VectorRagService

- vectorStore: VectorStore
- llamaChatClient: ChatClient

+ indexDocumentChunks(docGuid, text): List<Chunk>
+ chatWithDocument(docEntity, request): ChatResponse

BibTeXExtractionService

+ extractMetadata(fileName, sampleContent): BibTeXMetadata

StorageService

+ storeFile(file: MultipartFile, guid: String): Path
+ extractTextContent(path: Path): String
+ deletePhysicalAsset(guid): void

PACKAGE: com.personallibrary.repository

**<<interface>>
DocumentRepository**

+ findByGuid(guid: String): Optional<DocumentEntity>
+ findByPreviousVersionGuid(guid): List<DocumentEntity>

**<<CustomMongoRepository>>
DocumentRepositoryCustomImpl**

- mongoTemplate: MongoTemplate

+ searchDocuments(fileName, title, author, edition, format, content, pageable): Page<DocumentEntity>

persists / queries

PACKAGE: com.personallibrary.model

**<<Document (collection="documents")>>
DocumentEntity**

- id: String
- guid: String
- previousVersionGuid: String
- versionNumber: int
- fileName: String
- fileSize: long
- fileSizeFormatted: String
- format: String
- uploadDate: Instant
- editDate: Instant
- bibtex: BibTeXMetadata
- bibtexRaw: String
- summaries: Map<String, SummaryRecord>
- contentExcerpt: String
- fullContent: String
- chunks: List<DocumentChunk>
- versionHistory: List<DocumentVersionSnapshot>

**<<ValueObject>>
BibTeXMetadata**

- entryType: BibTeXType
- bibKey: String
- title: String
- author: String
- year: String
- journal: String
- booktitle: String
- volume: String
- publisher: String
- doi: String
- keywords: String

+ toRawBibTeX(): String
+ validateMandatoryFields(): boolean

SummaryRecord

- modelName: String
- modelKey: String
- summaryText: String
- createdAt: Instant
- durationSeconds: double
- durationFormatted: String

DocumentChunk

- id: String
- chunkIndex: int
- text: String
- embeddingVector: float[]

<<enum>> BibTeXType

ARTICLE, BOOK, INPROCEEDINGS
TECHREPORT, PHDTHESIS, MISC